

Chapter – 12

MINING

12.1 Introduction

Mining is the process of extracting valuable minerals or other geological materials from the earth.

Mining involves a number of stages which occur in a sequence. This sequence of stages is known as the mining sequence. The mining sequence covers all aspects of mining, including: prospecting for ore bodies, analysis of the profit potential of a proposed mine, extraction of the desired materials and, once a mine is closed, the restoration of all lands used for mining to their original state

To develop mines we have to understand the geology of the area.

The formation of ore body is important in this respect. An ore is considered to be an aggregation of minerals from which a metal or metallic compound can be recovered economically on a commercial scale. When the percentage of metal or valuable in the ore is too low for profitable recovery, the rock ceases to be an ore.

A mineral may be regarded as a naturally occurring chemical compound having a definite chemical composition and crystal structure. The physical properties of minerals play the most important role in the economic processing of various ores.

Physical properties of minerals:

The physical properties of minerals can be determined without the use of chemical tests. They depend upon the kind and arrangement of atoms in their crystal structures. The various physical properties of minerals include transparency, luster, colour, specific gravity, hardness, cleavage, fracture, magnetic properties, electrical properties, radioactive and optical properties.

Transparency

This term is used to describe the ease with which we can see through a mineral. Three terms for transparency are in common use i.e. opaque, transparent & translucent. Opaque minerals are those through which no light can be seen. Transparent minerals are those which can be seen through clearly. Translucent minerals are those through which a little light can be seen.

Luster

This may be defined as the amount and quality of the reflection of light from mineral surface. The luster of mineral refers to its surface appearance.

Colour

In most cases the colour of mineral is due to absorption of certain wavelength of light energy by the atoms making up the crystal. The remaining wavelengths of the light that are not absorbed give the sensation of colour to the eye.

Luminescence

This refers to the emission of light by a mineral which is not the direct result of incandescence. Luminescence in most minerals is faint and can be seen only in the dark. Minerals which luminesce during exposure to ultraviolet light and x-rays, are called fluorescent.

Specific Gravity

Specific gravity of a particular mineral is practically constant, it may vary a little with the presence of some impurities. The difference in specific gravity affords one of the surest means of separating minerals from each other and has been put to practical use. Simple washing in water affects an efficient separation of gold grains from quartz sand, whereas use of heavy liquids affects the separation of lighter coal from heavier shale. Difference in specific gravities forms the basis of a class of ore-dressing process known as 'gravity concentration method'.

Hardness

This may be defined as the ability of a mineral to resist scratching. This is different from the ease with which it can be broken. Diamond is one of the hardest material known, but it can be shattered easily. Like other physical properties, the hardness is dependent on the kinds and arrangements of atoms in mineral structures. The basis of the test is a set of minerals selected by the Austrian mineralogist, F. Mohs to which all other minerals could be compared to determine relative hardness. The scale numbered 1 to 10 in order of increasing order of hardness became known as Mohs Scale of Hardness, and is very useful as a means for identifying minerals or quickly determining hardness. The scale is not a linear scale, but somewhat arbitrary

Mohs scale of Hardness

Hardness	Mineral	Associations and Uses
1.	Talc	Talcum powder
2.	Gypsum	Plaster of paris. Gypsum is formed when seawater evaporates from the Earth's surface.
3.	Calcite	Limestone and most shells contain calcite.
4	Flourite	Fluorine in fluorite prevents tooth decay.
5	Apatite	Apatite is an inspirational stone. It develops psychic abilities and spiritual attunement. Use it to aid communication and self-expression. Apatite is most often seen in blue, but also can be found in brown, pink, yellow, green (from Spain called asparagus stone) and a rare variety of violet. Apatite heals bones, aids absorption of calcium, helps cartilage, bones, teeth and motor skills. Relieves arthritis, joint problems. Overcomes hypertension
6	Orthoclase Feldspar	Orthoclase is a feldspar, and in German, "feld" means "field".
7	Quartz	Quartz is the most common mineral found on the surface of the Earth. A significant component of many igneous, metamorphic and sedimentary rocks, this natural form of silicon dioxide is found in an impressive range of varieties and colours.
8	Topaz	The November birthstone. Emerald and aquamarine are varieties of beryl with a hardness of 8.
9.	Corundum	Sapphire and ruby are varieties of corundum. Twice as hard as topaz.
10	Diamond.	Used in jewelry and cutting tools. Four times as hard as corundum

Exploration

The purpose of exploration is to determine the thickness of ore body/coal seam and also to ascertain the quality of the ore/coal. For this purpose a bore hole is driven down the hole through core drilling and core is recovered which is placed in a core box with utmost care. The geologist/mining engineer measure the core length of the earth crust carefully and a picture is drawn on paper of the length and location of the ore body/coal seam. This process of keeping the core in core boxes systematically and maintaining a log book of the same with mentioning details of its physical properties as instant is known as logging.

Simultaneously ore/coal seam sent to laboratory for quality analysis. This provides the basic information for the Geological and Assay database for any mining software like surpac, datamine to be used for the Reserve estimation, establishing average quality, Ore modeling, preparation of the mine plan and finally the mine scheduling to produce targeted quantity of Ore with stipulated quality.

Bench wise excavation plan

To meet the targeted production with desired quality a systematic and scientific bench wise excavation plan is prepared synchronizing the overburden removal to expose sufficient ore benches for the desired production and excavation from different ore benches of different ore quality which can be blended at the crusher level for the desired final product for dispatches to the steel plant. Bench parameter like its height, width and gradient is maintained as per DGMS permission.

Reading of excavation plan

The size and shape of topography where mining is to be carried out is depicted in a mine surface plan which is drawn on a suitable scale together with every surface features in a schedule manner called mine surface plan. The various surface features are shown according to a specified schedule under the statute. Excavation plan can be prepared on the mine surface plan or separately on mine working plan showing all the mine working benches/faces clearly demarcating the area with the length and breadth of the planned excavation in particular stipulated period. Normally excavation plan is prepared every month with respect to the annual excavation plan under five year Mining Scheme approved by Indian Bureau of Mines. Excavation plan can vary with deviations in the requirements, but annual planned quantity shall not be increased more than the quantity stipulated in the mining scheme without the permissions of IBM.

Short & Long term excavation plan

Long term plans

All the long term excavation plans have to be made in line with the mines plans, mining scheme and the mine closure/progressive mine closure plans approved by IBM.

Progressive Mine Closure Plan is the essence of mine planning in which the planning starts from the final pit limit at the end of mine life with systematic reclamation and rehabilitation so that at the end it is restored to the original landform as far as possible.

Mine Plan is the plan of systematic excavation plan for the applied lease period normally for 20 years with due considerations of the Environmental protection and the mineral Conservation. There are always possibilities of the deviations from the mine plan if there are shortages from the target or any changes in demands.

Mining Schemes;- To accommodate the deviations from the original mine plan a Mining Scheme is prepared after every 5 years of the Mine Plan period.. Five year mining scheme shows the modified mine excavation plan with clear details of each year excavation. All the plans are prepared for mining of ore body/coal keeping in view the objective and mission of our Company. In long term plan Govt. of India five years plan in the development of society

& our Country are also taken care of. For example vision 2025 have been drawn where steel production is augmented to 100 million tones in India.

Short term plans

In order to reach the goal and to fulfill the objective of the Organization based on the long term excavation plan like the mine plan and the mining schemes, short term plan is prepared for annual/monthly excavations where every detail is worked out.

Waste dump/making of waste dump plan/slope stability/illumination

Waste dump is defined as stacking of broken waste material which do not conform to economic value but the volume is so high that a careful assessment and planning of waste dump is required which plays a vital role in running an open cast mine. It should be stacked on non mineralized area on a firm ground or old mine working within the lease. It should be so planned that re-handling of waste dump is minimum or not at all required i.e. away from ore body/coal. With the present environment concerns dumping in the hill slopes is not being permitted by the forest as well as the pollution control departments. It is therefore desired that mine excavation should be planned in such a way that excavated mine working is available for back filling of the old mined out area from where entire mineral has been exploited.

. The waste dump can be made in different segment in 30m height so that slope failure is minimum or not at all possible. For this purpose a horizontal length all along the periphery is left. The draw angle i.e. the angle of repose of waste material is normally kept at 33.5 degree from vertical to avoid slope failure. Waste dump are being manned by mining sirder /mining mate & dump man and frequented by shift I/c. With the help of dozer/pay loader a berm should be prepared all along the dump yard in the direction of extension of dump. Regular water sprinkling to suppress the dust should be done. While designing a dump utmost care should be taken that the waste is not carried away by the rain water into the natural water course or the river stream during monsoon.

While breaking a new area if any top soil is available it should not be dumped with the waste materials at the waste dump but it should be stacked separately for the future use for plantation in the process of mine rehabilitation.

If the mine operation continue in the dark hours the dump shall be adequately lighted to facilitate the free flowing movement of dumpers. Minimum illumination level as per Mines Regulation, illumination to the level of 3.5 lux is required to be maintained.

Equipment planning/Selection of equipment

The selection of equipment for opencast mine is dependent on the size and shape of ore body, ore reserves, volume of production desired and the time frame to close the mine. For example if a deposit is highly erratic and selective mining is required, it has to be worked manually. In a small deposits higher capacity of equipments are not desired whereas in big deposits where large excavation is required, higher capacity equipments are effective and economical in operation. So selection of equipment is an important phenomenon for open cast mine. The following combinations are suitable and normally used as per the rocks/minerals to be excavated.

- a) Excavator dumper combination
- b) In pit crusher belt conveyor combination
- c) Bucket wheel excavator

d) Drag line

Bucket wheel excavator is a large capacity continuous excavator which is used in soft formations for directly excavating material from the mine faces and transporting to the desired locations through the conveyors. It is also being used in some mines and the ore handling plants at the steel plants for reclamation of the lumps/fines of the Iron ore.

Dragline is a giant excavator which can be used for directly side casting the overburden to large distances. It can be effectively used also for temporary shifting of the overburden for mining and backfilling the mined out area after extraction of the entire deposit where the thickness of the deposit are not very thick..

Equipment

Equipments required for opencast mining :

- a) Excavators : Rope Shovel / Hydraulic Excavator (Front end or Back hoe)
- b) Dumper: 35t capacity, 50t capacity, 85t capacity, 100t, 120t capacity etc.
- c) Tipper / hyva : 16Te/25Te
- d) Drill : Electric driven, Diesel driven
- d) Dozer: D-155,D-355,D-510
- e) Grader
- f) Water sprinkler

Major Manufactures of Earth Moving Equipment

BEML, Caterpillar, TELCON, L&T – Komatsu, AtlasCopco etc

Combination of equipment

Combination of equipment is dependent on the volume of production required with a targeted quality. In all the SAIL mines except the manual mines, excavator-dumper combination is used for excavation and the transport of ore to the crushing and the screening plant large dia drills from 100mm to 165 mm are used for drilling. Dozers and the road graders are used for leveling at the mines and the mines roads respectively. Big water sprinklers are used to suppress the dust in the mines which are very essential because no other operation in the mine is possible unless the dust is suppressed.

12.2 Mines Operation

Drilling/Placement of drill/Sub grade drilling

Ahead of the mine production faces in different benches, a drilling block should be maintained of sufficient length and width in all the faces to manage quantity of material at least for one week. This enables easy weekly blasting and prevents from frequent shifting of the machines. For good blasting efficiency effective drilling parameters should be followed. Drilling parameters are dependent on the rock characteristics. Distance from the free face to the first line of the holes is known as Burden, hole to hole distance in a row is called spacing and the extra drilling more than the height of the bench is called the sub-grade drilling. Drilling block is properly leveled with the use of a dozer and prior to start of the drilling operations, positions of the drill holes are physically marked on the ground as per the desired drilling parameters. Drilling parameters can be fixed on the trial and error basis following the general prescribed thumb rules. Harder rock requires smaller burden and spacing while the larger spacing and burden can be given in the soft rocks where less energy is required for breaking of the rocks.

. The following thumb rule is in practice for determining the drilling parameters in the mine

- a) The burden of drill holes can be 200 to 300 times the dia of the holes.
- b) Spacing can be 1.2 times of burden.
- c) Sub-grade drilling should be approx 10% of the bench height.

Marking of holes on the surface can be on a square pattern or the staggered pattern. When the spacing of the holes in the second row are in line with the first row making a square, it is called the square pattern and when the holes of the second row are placed between the two holes of the first row forming a triangle are called staggered pattern. Sub grade drilling are required to compensate the drilling loss after the drill is removed, and maintained the uniform bench floor. For good blasting, length of the hole should be at least 10% greater than the height of the bench or otherwise toes will be formed at the foot of the bench which restricts the digging efficiency of the excavator

Blasting

Blasting is done to provide fragmented rock mass for shoveling. It's the lifeline of a mines operation because the efficiency of all other operations like excavation, loading of dumpers, transportation and crushing and the screening are dependent on the blasting efficiency i.e. the fragmentation of the blasted mass. Different types of explosives are used for rock fragmentation.

- a) High prime charge
- b) Prime charge
- c) Base charge
- d) Column charge
- e) Bulk Emulsion Explosive - Site Mixed Emulsion (SME) Explosive

Explosive :

An explosive has four basic characteristics: (1) It is a chemical compound or mixture ignited by heat, shock, impact, friction, or a combination of these conditions; (2) Upon ignition, it decomposes rapidly in a detonation; (3) There is a rapid release of heat and large quantities of high-pressure gases that expand rapidly with sufficient force to overcome confining forces; and (4) The energy released by the detonation of explosives produces four basic effects; (a) rock fragmentation; (b) rock displacement; (c) ground vibration; and (d) air blast.

Site mixed explosive

It is a new and advance concept of an on-site mixed emulsion explosive over the ANFO or other packaged explosive. It is characterized in that, by percentage to the quality, comprises following component composition: 60 ~ 78% ammonium nitrate, 16 ~ 30% water, 1 ~ 2.5% emulsifying agent, 4 ~ 7% microemulsified oil phases and 0.1 ~ 0.5% sensitizing agent.

Advantages of bulk emulsion over ANFO or other packaged explosive products :

- Easier transport, handling and storage.
- Universal, detonator sensitive explosive.

- Accurate explosive consumption count.
- Shorter charge times.
- Improved work environment.
- Increased VoD.
- Full coupling.
- Excellent water resistance.

Blasting Circuits in Open Pit Mines-

Blasting is a critical operation in open pit mining for fragmenting rock and facilitating its excavation, loading, and transportation. The effectiveness of blasting largely depends on the design of the blasting circuit, which determines the sequence and timing in which blast holes are initiated. Over the years, different blasting systems have been used in open pit mines, including plain detonators with detonating fuse, electric detonators, and non-electric detonators. Recent regulatory developments have also encouraged the transition to more advanced electronic detonator systems.

In the conventional method using **plain detonators**, the blasting circuit is generally connected with detonating fuse. Detonating relays are placed between rows of blast holes to provide row-to-row delay, typically around 17/25/42 milliseconds, which helps in achieving controlled blasting. These delays allow the front row of holes to break first and create a free face for the subsequent rows. Providing a free face is essential in blasting technology because it enables the rock to move and break efficiently, leading to improved fragmentation compared to simultaneous or solid blasting without delays.

Another traditional system involves the use of **electric detonators**. In this system, delay sequencing between rows is achieved through electric delay detonators connected in series and initiated by an exploder. Electric circuits allow controlled initiation of multiple blast holes, but they are susceptible to hazards associated with stray electrical currents and require careful handling and circuit testing before firing.

With technological advancements, the **non-electric detonator (NONEL) system**, based on shock tube technology, has become widely adopted in large open pit mines. In this system, instead of detonating fuse or electric wires, a hollow plastic shock tube transmits the initiation signal to the detonator. The tube contains a reactive explosive coating on its inner surface that propagates a low-energy signal when ignited. This signal travels through the tube without producing significant external disturbance, making the system largely immune to stray electrical currents. In the NONEL system, delays can be arranged hole-to-hole rather than only row-to-row, creating additional free faces during the blast and significantly improving rock fragmentation.

The NONEL system offers several advantages. It preserves the explosive energy because the shock tube does not destroy the bubble energy of explosives, unlike detonating fuse where the flame directly contacts the explosive column. This leads to better utilization of explosive energy in breaking the rock rather than losing it in the form of air blast and ground vibrations. Hole-to-hole delays also produce improved fragmentation, reduced rock throw, and better muck pile formation. Additionally, the system results in lower vibration levels, improved safety in handling and blasting operations, and allows a large number of blast holes to be fired effectively in a single shot. Since the surface connections are completed before the down-the-hole delays are initiated, the chances of misfire are comparatively lower.

In recent years, regulatory authorities have encouraged the adoption of more advanced initiation systems. As per the guidelines of the **Petroleum and Explosives Safety Organisation (PESO)**, the use of electric detonators is being phased out, with their elimination targeted before **30 June 2025**. Mines are therefore transitioning toward

electronic detonator systems, which offer highly precise delay timing, improved control over blasting outcomes, and enhanced safety.

Thus, blasting circuits in open pit mines have evolved from simple detonating fuse systems to sophisticated non-electric and electronic initiation technologies, aiming to achieve better fragmentation, improved safety, reduced environmental impact, and higher operational efficiency.

Advantages of using nonel technology :

- a) Nonel tubes do not destroy the bubble energy of explosives whereas in detonating fuse the flames traveling is in contact with explosives destroying its bubble energy. In this way by saving the explosive energy it leads to cost saving and better fragmentation. More energy is utilized in breaking the rock instead it is wasted in the atmosphere causing air blast and the blast vibrations.
- b) Each hole gets blast at different timing thereby extra free face is created leading to better fragmentation
- c) Less throw and better muck pile
- d) Less vibration
- e) Safer in handling and blasting.
- f) Large no. of holes can be blasted effectively in single shot providing large volume of the blasted mass.
- g) There are less chances of misfire as all the surface connections are charged before down the hole non electric delays are initiated.

Site Mixed Emulsion (SME) Explosive :

Compared with a conventional on-site mixed loading emulsion explosive, the on-site mixed loading emulsion explosive provided by the invention can significantly improve explosion properties of the on-site mixed loading emulsion explosive, at the same time, reduce production costs, moreover, improves transportation safety of the oil phase material, and is particularly suitable for industrialized production.

Operation schedule/deployment/monitoring

Operation schedule is a daily/weekly operation plan taking into consideration of quality parameters & targeted production. It is synchronization between available bench wise blasted material quality, quantity and available shovel combination. To get a desired result the bench wise shovel operation must be monitored throughout the shift.

Transport /arrangement of transport

For transporting men and material, arrangement of transport is vital in a mines operation.

Quality/monitoring

It's a desirable practice that wining of ore shall commence after blasting of ore body. Before blasting all extraneous materials shall be removed. Bench wise quality excavation is monitored by scheduling deployment of shovels in a shift. The shift operation should follow the plan for getting the desired quality.

Crushing plant

Transported ore from the quarry is fed to primary crusher. Primary crusher is normally a gyratory crusher and sometimes Jaw crushers, which crush the ore upto a size of 300mm. The crushed ore is then feed to a secondary cone crusher through conveyor belt to get a desired size of 40mm and below. The mixture of sized lump & fines is stocked into a stock pile for feeding to screening plant through conveyors.

Screening plant

The job of screening plant is to separate lump (-40mm to +10mm) & fines (-10mm) through double deck screen and transport the material to Ore Handling Plant stockpiles through conveyors. In wet screening the lump & fines are washed with waters in the double deck screen for removing impurities during monsoon.

Iron Ore Beneficiation Plant:-

To get the desired quality of iron ore by removing silica & alumina from ROM, It is further beneficiated to recover the iron ore from the slime/tailing of crushing unit. For this the tailings or slime has been passed and processed through slime beneficiation unit or silica reduction unit. In these plant, as a main equipment Hydro-cyclones and Magnetic Separators are being used to recover iron ore fines of high grade.

Ore Handling Plant

The job of ore handling plant is to feed the wagon loader from stock pile by a bucket wheel excavator. From the bucket wheel excavator the lumps & fines are transported to wagon loader for loading into wagons by conveyors.

Loading and dispatch

Managing loading and dispatch within the free time without incurring demurrages and loading correct weight to avoid heavy penal freight is the prime requirement and the big challenge. There are different systems of loading in different mines. Loading are done by wagon loader, shovels or the pay loaders. No. of the loading equipments depend on it's capacity and the free time provided by the railways.

12.3 Safety in Mines

Safety / Personal Protective Equipment (PPE)/Safety of men/machine

The safety of persons employed in mines and also the equipment are the responsibility of the statutory persons specially deployed as per guidelines of DGMS viz. Mining Mate/ Foreman / Electrical supervisors / Astt. manager / Mines Manager & other engineers working in the mine. There are hazards associated with blasting, transportation of coal/ore, use of electrical energy. As a general rule all persons working in mines must wear personal protective equipments, follow general principles laid down in permission letters by the

Directorate General of Mines Safety Official as per MMR 1961/CMR2017. General condition of mines atmosphere shall be congenial, airborne dust shall be suppressed at the place of formation by water sprinkling.

All Equipment's like Excavator, Dozer, Drill etc. shall be kept at a safe distance during blasting to avoid damage from flying fragments.
